

Provenance-Aware Memory Architecture for Chronic Healthcare Continuity

Continuity Burden, Provider Fragmentation, and the Narrative Gap in Longitudinal Care

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Abstract

Chronic illness affects over 60% of adults in the United States and accounts for the majority of healthcare expenditure, yet the systems designed to support chronic disease management remain episodic, provider-centered, and fundamentally stateless from the patient's perspective. Patients with multiple chronic conditions see an average of five or more specialists per year, and each encounter typically begins with the patient re-establishing medical history, symptom context, treatment responses, and personal circumstances—a process this paper terms *healthcare continuity burden*. This paper extends the continuity burden framework developed in prior work on spaceflight and extraterrestrial habitation to the domain of chronic healthcare, arguing that the patient's repeated re-disclosure of medical and personal context across fragmented provider interactions constitutes a measurable cost to both care quality and patient well-being. Drawing on the Circular Associative Memory Architecture (CAMA), this paper proposes a patient-centered persistent memory system that retains longitudinal context across provider transitions, tracks symptom trajectories and treatment responses over timescales that exceed any single provider relationship, and bridges the *narrative gap*—the absence of contextual details about symptom evolution, lifestyle factors, and treatment adherence—that limits clinicians' ability to deliver personalized care. The paper identifies five healthcare-specific failure modes of stateless AI systems, proposes a patient-sovereign data governance model, and presents a worked scenario demonstrating longitudinal intervention capability across provider transitions. This is the third paper in a four-part series examining the societal implications of persistent memory architecture.

Keywords: chronic illness, healthcare continuity, continuity burden, provider fragmentation, narrative gap, CAMA, persistent memory, patient-centered care, longitudinal health tracking, lupus, autoimmune disease, provenance-aware memory, treatment burden

1. Introduction

Chronic diseases—including cardiovascular disease, diabetes, autoimmune disorders, chronic kidney disease, chronic respiratory conditions, and cancer—are the leading cause of death and disability globally, affecting approximately 60% of American adults and accounting for 90% of the \$4.5 trillion in annual U.S. healthcare expenditure [1, 2]. Unlike acute conditions with clear diagnostic and treatment arcs, chronic illness is characterized by indefinite duration, fluctuating severity, multi-system involvement, and the need for ongoing management that spans years or decades. The patient, not the provider, is the primary manager of their condition: monitoring symptoms daily, implementing treatment protocols, adapting to changing disease activity, and making moment-to-moment decisions about their own care [3].

Despite this reality, healthcare delivery systems remain designed around the episodic encounter. A patient with lupus sees a rheumatologist quarterly, a nephrologist for kidney involvement, a dermatologist for skin manifestations, a primary care provider for overall management, and potentially a psychiatrist for the psychological burden of living with an unpredictable autoimmune disease. Each provider has a partial view of the patient’s condition. Each encounter begins with some version of the question: “So, what brings you in today?”—even when the answer is the same disease the patient has been living with for years.

This paper argues that the patient’s experience of repeatedly re-establishing context across fragmented provider interactions constitutes a specific, measurable form of *continuity burden*—the same construct introduced in prior papers in this series to describe the cost of stateless interaction in spaceflight [4] and permanent habitation [5] contexts. In healthcare, continuity burden is not imposed by communication delay or geographic isolation but by a delivery system that treats each encounter as independent, each provider as a silo, and each patient as a stranger until proven otherwise.

Emerging research has identified a *narrative gap* in healthcare data: the absence of contextual details about symptom evolution, lifestyle factors, and treatment adherence that limits clinicians’ ability to deliver personalized care [6]. Existing AI applications for chronic disease management focus predominantly on diagnostic prediction, symptom monitoring, medication adherence, and wearable-based physiological tracking [7, 8]. To our knowledge, no current AI system addresses

the narrative gap through persistent, provenance-aware memory that retains the patient’s longitudinal story—not just their data points—across provider transitions and over timescales that exceed any single clinical relationship.

2. Healthcare Continuity Burden

2.1 Defining Healthcare Continuity Burden

Healthcare continuity burden is defined as the cumulative cognitive, emotional, and temporal cost imposed on patients who must repeatedly re-establish their medical history, symptom context, treatment responses, and personal circumstances across fragmented provider interactions. It is the healthcare-specific instantiation of the continuity burden construct introduced in the spaceflight context [4], extended from individual-to-system interaction to individual-to-fragmented-system interaction.

The mechanism operates at multiple levels. At the informational level, the patient must recall and communicate a complex medical history accurately under time pressure, often to a provider who has not read their chart or whose electronic health record contains only structured data fields that strip context from the patient’s narrative. At the emotional level, each re-disclosure of chronic illness—its limitations, its unpredictability, its impact on daily life—carries a cost, particularly when the patient perceives that the provider does not have the time or the system capacity to retain what they share. At the relational level, trust must be rebuilt with each new provider, and the patient’s sense of being known—of being seen as a whole person rather than a problem list—is reset at every transition.

For patients with complex, multi-system chronic conditions, healthcare continuity burden is not occasional; it is structural. It is built into the architecture of the delivery system.

2.2 Failure Modes of Stateless Systems in Chronic Healthcare

AI systems designed for chronic disease management inherit the statelessness of the broader healthcare architecture unless they are specifically designed to overcome it. Five healthcare-specific failure modes emerge:

Provider transition knowledge loss. When a patient changes providers—due to insurance changes, provider relocation, retirement, or dissatisfaction—the accumulated context of the prior relationship is largely lost. The new provider inherits a chart, not a relationship. Treatment decisions that were made in context (“we tried methotrexate first because the patient expressed

concern about the side effect profile of biologics”) appear in the record as bare prescriptions without the reasoning that informed them.

Symptom trajectory fragmentation. Chronic illness symptoms fluctuate over timescales—weeks, months, seasons, years—that no single provider encounter captures. A patient with lupus may experience a flare pattern that correlates with seasonal changes, stress cycles, or medication adjustments made months earlier. If the AI system supporting their self-management resets between sessions or does not track longitudinal symptom trajectories, these patterns remain invisible until the patient identifies them through their own observation—if they can.

Treatment response amnesia. Over the course of chronic illness management, patients may try dozens of medications, dosages, combinations, and non-pharmacological interventions. The record of what was tried, why it was discontinued, what worked partially, and what produced intolerable side effects is often scattered across multiple provider records, pharmacy databases, and the patient’s own memory. A stateless AI system cannot help a patient and provider avoid re-trying a medication that was already tried and failed three years ago with a different provider.

Contextual self-management erosion. Effective chronic disease self-management requires the patient to make daily decisions informed by their accumulated understanding of their own condition. A memory-aware AI companion can reinforce and extend this self-knowledge by tracking what the patient has learned about their own patterns, what strategies they have developed, and what contextual factors (sleep, diet, stress, weather, menstrual cycle) have correlated with symptom changes. A stateless system cannot build on prior self-management insights; it can only offer generic guidance that does not account for what the patient already knows about their own body.

Emotional withdrawal from support systems. The same disclosure fatigue documented in spaceflight contexts [4] operates in healthcare: patients who repeatedly experience their context being forgotten—by providers, by systems, by AI tools—eventually stop investing in disclosure. They shorten their answers, omit relevant history, and resign themselves to receiving care that does not account for their full situation. This withdrawal is rational from the patient’s perspective: the cost of thorough disclosure exceeds the expected benefit when the system demonstrates no capacity to retain it.

Table 1. Healthcare-Specific Failure Modes of Stateless AI Systems

Failure Mode	Patient Impact	Stateless System	Memory-Aware System
Provider transition	Treatment reasoning and relational	No retention of	Retains longitudinal

knowledge loss	context lost at each provider change	prior provider relationship context	treatment history with reasoning and patient preferences
Symptom trajectory fragmentation	Long-term symptom patterns invisible to episodic encounters	Each session independent; no trajectory analysis	Tracks symptom patterns across months and years
Treatment response amnesia	Previously tried and failed treatments re-attempted; side effects re-experienced	No memory of prior treatment responses	Complete treatment history with outcomes, side effects, and discontinuation reasons
Self-management erosion	Patient's accumulated self-knowledge not reinforced or extended	Generic guidance without patient-specific context	Builds on patient's own pattern recognition and self-management strategies
Emotional withdrawal	Patient stops investing in disclosure; care quality degrades silently	Cannot detect or address withdrawal driven by disclosure fatigue	Tracks engagement patterns; reduces disclosure cost through retained context

Collectively, these failure modes constitute *healthcare continuity burden*: the cost borne by the patient and the care system when longitudinal context must be reconstructed at every encounter. Unlike institutional continuity burden in settlement contexts [5], healthcare continuity burden is distributed across a fragmented landscape of providers, systems, and temporal gaps rather than concentrated at periodic rotation boundaries.

3. A Patient-Centered Persistent Memory System

3.1 Architecture Adaptation for Healthcare

The CAMA architecture, described in prior work [9, 10] and available as an open-source implementation at <https://github.com/LoriensLibrary/cama>, provides the foundational three-layer design (archive, relational index, active ring buffer) with provenance tracking. In the healthcare context, the architecture is adapted with two key principles:

Patient sovereignty over data. Unlike the spaceflight and habitation contexts where the AI system serves both the individual and the institution (mission control, settlement operations), in

healthcare the primary client is the patient. The memory system belongs to the patient, travels with the patient across providers, and is governed by the patient's consent decisions. This is a fundamental architectural choice: the system is not a provider tool with patient data; it is a patient tool that can share data with providers when the patient authorizes it.

Narrative preservation alongside structured data. Electronic health records capture structured data—diagnosis codes, lab values, prescription lists. They do not capture the patient's narrative: why a particular treatment was chosen, how the patient experienced a side effect, what the patient has learned about their own flare patterns, what non-medical factors (housing instability, caregiving responsibilities, financial stress) affect their ability to manage their condition. CAMA's emotional keying and provenance tracking enable storage of both structured health data and narrative context, preserving the information that the patient currently carries in their own memory and must re-communicate at every encounter.

3.2 Healthcare-Specific Capabilities

Longitudinal symptom tracking with contextual annotation. The system tracks not just symptom occurrence and severity but the patient's own understanding of context: "My joints are worse this week—I think it's the weather change, but it could also be that I ran out of my anti-inflammatory three days ago." Both the symptom report and the contextual annotation are stored with provenance, enabling the system to build a richer model of the patient's condition than structured data alone can provide.

Treatment history with reasoning and outcomes. Every medication, dosage change, and non-pharmacological intervention is stored with: (1) who prescribed it, (2) why it was prescribed, (3) how the patient responded, (4) why it was continued, modified, or discontinued, and (5) the patient's subjective experience. This history travels with the patient across providers. When a new rheumatologist asks "Have you tried hydroxychloroquine?" the system can respond: "Yes—prescribed by Dr. Martinez in 2022, discontinued after four months due to visual disturbance side effects. Patient reports it was partially effective for joint symptoms before discontinuation."

Provider transition briefing. When a patient begins seeing a new provider, the memory system can generate a structured briefing that goes beyond the standard chart transfer: a narrative-rich summary of the patient's condition history, treatment trajectory, self-management strategies, known triggers, and the patient's own priorities and concerns. This briefing is generated from the patient's perspective and authorized by the patient, giving the new provider context that would otherwise take months of relationship-building to accumulate.

Self-management reinforcement. The system can reflect the patient’s own pattern recognition back to them: “You’ve noticed that your fatigue tends to worsen about a week before a flare. Your last two flares were preceded by the same pattern. Would you like to discuss preventive measures with your provider?” This capability builds on what the patient already knows rather than offering generic guidance, transforming the AI from an information source into a longitudinal self-management partner.

3.3 Worked Example: Lupus Management Across Provider Transitions

The following scenario is illustrative rather than derived from clinical trial data. It demonstrates how a CAMA-equipped patient companion manages continuity across provider transitions for a patient with systemic lupus erythematosus (SLE).

Year 1. Patient Maria, 34, is diagnosed with SLE by rheumatologist Dr. Chen. The AI companion stores the diagnosis, initial lab values, and Maria’s emotional response to the diagnosis. Over the next twelve months, it tracks her responses to hydroxychloroquine (effective for joint pain, discontinued at month 4 due to visual symptoms), mycophenolate (started month 5, effective for kidney involvement, ongoing), and prednisone tapers (three attempts, two successful, one resulting in flare). It also stores Maria’s self-reported patterns: fatigue worsens with heat exposure, joint pain correlates with sleep quality, and stress at work reliably precedes flares by 5–7 days.

Year 2. Maria’s insurance changes. She must find a new rheumatologist. Dr. Chen’s records are transferred as a standard chart, containing diagnosis codes, lab results, and prescription history. In a stateless system, Maria’s new rheumatologist, Dr. Okafor, reviews the chart and begins the relationship with: “So, tell me about your lupus.” Maria must reconstruct two years of lived experience in a 30-minute intake appointment.

In the CAMA-equipped scenario, Maria authorizes her AI companion to generate a provider transition briefing for Dr. Okafor. The briefing includes: *SLE diagnosed January 2024 by Dr. Chen. Kidney involvement (Class III nephritis, biopsy-confirmed). Hydroxychloroquine tried and discontinued due to visual disturbance (months 1–4). Currently on mycophenolate 1000mg BID with good renal response. Three prednisone tapers attempted; patient is steroid-sensitive and flares reliably during taper below 7.5mg. Known triggers: heat, sleep disruption, work stress (5–7 day lag). Patient’s priority: minimizing steroid dependence. Patient’s concern: kidney function trajectory.* Dr. Okafor receives in five minutes what would otherwise take months to learn. The relationship begins with context, not from zero.

Year 3. Maria’s kidney function shows gradual decline. The AI system, tracking her lab results longitudinally, identifies a trend that no single quarterly lab value would flag: GFR has decreased by 8 points over 14 months, with each individual measurement falling within “normal” range but the trajectory indicating concern. The system flags this for discussion at her next appointment: “Your kidney function tests have been individually normal, but I’ve noticed a gradual downward trend over the past year. You may want to discuss this trajectory with Dr. Okafor.” In a stateless system, this trend would not be detected until it crossed a clinical threshold—potentially months or years later.

4. Patient-Sovereign Data Governance

Healthcare data governance for a patient-centered memory system differs fundamentally from the institutional models proposed in Papers 1 and 2. In spaceflight, the institution (mission control) has a legitimate operational interest in crew health data. In healthcare, the patient is the sovereign owner of their own narrative, and any sharing of that narrative requires explicit, informed, revocable consent.

Tier 1: Patient-only (default). All memories are stored under the patient’s control. The AI companion interacts with the patient, stores their disclosures, tracks their patterns, and builds their longitudinal model. No information leaves the patient’s memory system without explicit authorization.

Tier 2: Provider-shared (patient-authorized). The patient can authorize specific memories or generated summaries to be shared with specific providers. Authorization is granular: a patient can share their treatment history with a new rheumatologist without sharing their mental health disclosures. Authorization is revocable: a patient can withdraw a provider’s access at any time.

Tier 3: Research-shared (patient-authorized, de-identified). With explicit consent, patients can contribute de-identified longitudinal data to research repositories. The provenance-aware architecture enables meaningful de-identification because the system can distinguish between factual medical data (shareable after de-identification) and personal narrative context (requiring more careful handling).

This governance model inverts the current healthcare data paradigm. Rather than institutions holding patient data and patients requesting access, the patient holds their own longitudinal memory and providers request access. This inversion aligns with emerging patient data sovereignty movements and addresses a fundamental power asymmetry in current healthcare information systems [11].

5. Safety Considerations

The safety concerns identified in Papers 1 and 2 (false-memory persistence, retrieval-induced amplification, adversarial insertion) apply to healthcare with additional domain-specific risks:

Medical misinformation persistence. If a patient stores inaccurate medical beliefs (“My naturopath said lupus is caused by gluten”), the system must not amplify these beliefs as established medical facts. Provenance tracking distinguishes between patient-reported beliefs, provider-communicated medical information, and system-generated inferences. The system should not contradict the patient’s beliefs unprompted, but when a provider-sourced memory contradicts a patient-reported belief, the discrepancy should be flagged for the patient’s awareness.

Liability and clinical decision-making. A memory-aware AI companion is not a diagnostic tool, and its longitudinal observations—however accurate—do not constitute medical advice. The system’s role is to surface patterns for the patient and, with authorization, for their providers. Clinical decisions remain the responsibility of licensed providers. The system must communicate this boundary clearly and consistently, particularly when it surfaces concerning trends like the GFR trajectory in the worked example.

Over-dependence and self-diagnosis risk. A system that tracks patterns effectively may lead patients to over-interpret fluctuations, self-diagnose, or modify treatment without provider consultation. The system should be designed to encourage provider engagement rather than replace it—surfacing patterns as questions to discuss with providers rather than as conclusions to act on independently.

Access equity. A patient-centered persistent memory system must not become a tool that benefits only technologically literate or well-resourced patients. If persistent memory improves care quality for patients who have it, its absence for patients who do not have it becomes a health equity issue. This paper does not resolve this concern but identifies it as a critical design and policy consideration for any implementation pathway.

6. Evaluation Framework

Evaluating persistent memory for chronic healthcare requires a fundamentally different design than the analog-trial approaches proposed for spaceflight and habitation. Healthcare evaluation must account for real-world provider fragmentation, longitudinal timescales, and patient diversity.

Study design: A longitudinal cohort study enrolling patients with multi-provider chronic conditions (e.g., SLE, rheumatoid arthritis, type 1 diabetes, multiple sclerosis). Participants are

randomized to: Condition A (standard care with no AI companion), Condition B (stateless AI health companion), or Condition C (CAMA-equipped AI companion with persistent memory). Minimum follow-up: 12 months, with at least one provider transition during the study period.

Primary outcomes: (1) Patient-reported continuity burden, measured by a healthcare-adapted version of the Continuity Burden Inventory (CBI-H) administered quarterly. (2) Provider-reported contextual preparedness, measured by a brief survey administered to providers at the first encounter after patient enrollment. (3) Longitudinal pattern detection, measured by comparing the number and timing of clinically significant trends identified by each condition.

Secondary outcomes: Patient engagement with the AI system, disclosure depth and frequency, treatment history accuracy (compared against pharmacy and medical records), and patient-reported satisfaction with care continuity.

7. Limitations

This paper is conceptual and architectural. No clinical trial data supports the proposed system, and the worked example is illustrative. The healthcare-adapted Continuity Burden Inventory (CBI-H) has not been developed or validated. The patient-sovereign data governance model proposed here faces significant regulatory, technical, and interoperability challenges (including HIPAA compliance, integration with existing EHR systems, and cross-platform data portability) that are identified but not resolved. The CAMA prototype is currently a single-user system without healthcare-specific adaptations. The access equity concern identified in Section 5 is a structural limitation that cannot be resolved through architecture alone. These limitations should be understood as defining a research and development agenda, not as evidence of deployed effectiveness.

8. Broader Implications and Future Work

This paper brings the persistent memory framework from extraterrestrial environments to the largest affected population: the hundreds of millions of people worldwide living with chronic conditions. The continuity burden they experience is lower in intensity than what astronauts face but vastly larger in scale and duration. A patient with lupus diagnosed at age 30 may manage their condition for fifty years across dozens of provider relationships. The cumulative continuity burden of that trajectory is immense and almost entirely invisible in current healthcare discourse.

The final paper in this series (Paper 4: Haven) examines the most personal application of persistent memory: psychological support for veterans through music-mediated emotional continuity. Where

this paper addresses the fragmentation of medical knowledge, Paper 4 addresses the fragmentation of emotional knowledge—the challenge of supporting populations who carry experiences they cannot or will not articulate through conventional therapeutic language.

Across all four papers in this series, the argument converges: persistent, provenance-aware memory is not a feature of AI systems. It is infrastructure. Its absence is not a neutral design choice; it is a decision to impose continuity burden on the populations who can least afford it—isolated astronauts, rotating settlement crews, chronically ill patients, and veterans carrying invisible wounds.

9. Conclusion

The healthcare system asks chronically ill patients to be the carriers of their own longitudinal story, then provides them with tools that cannot remember what they said yesterday. Electronic health records capture data points but not narratives. Provider transitions erase relational context. AI health tools offer generic guidance to patients whose conditions are anything but generic. The result is a system in which the most knowledgeable person in the room—the patient—bears the highest continuity burden.

A patient-centered persistent memory system does not replace providers, override clinical judgment, or solve the structural problems of healthcare fragmentation. What it does is ensure that the patient’s longitudinal story—their treatment history, their symptom patterns, their self-management wisdom, their priorities, and their voice—persists across the discontinuities of a system that was not designed for them. That persistence is not a luxury. For the 60% of Americans living with chronic conditions, it is a fundamental gap in their care infrastructure.

References

- [1] Hacker, K. (2024). The burden of chronic disease. *Mayo Clinic Proceedings: Innovations, Quality & Outcomes*, 8(1), 112–119.
- [2] Centers for Disease Control and Prevention. (2024). About chronic diseases. CDC.
- [3] Bodenheimer, T., Lorig, K., Holman, H., & Grumbach, K. (2002). Patient self-management of chronic disease in primary care. *JAMA*, 288(19), 2469–2475.
- [4] Reinhold, A. (2026). Persistent memory as mission-critical infrastructure for long-duration spaceflight. Zenodo. DOI: 10.5281/zenodo.19257809.
- [5] Reinhold, A. (2026). Memory-aware AI systems for permanent lunar and Martian habitation. Zenodo. DOI: 10.5281/zenodo.19260574.

- [6] Moëll, B. & Sand Aronsson, F. (2025). Journaling with large language models: A novel UX paradigm for AI-driven personal health management. *Frontiers in Artificial Intelligence*, 8, 1567580.
- [7] Hwang, M., Zheng, Y., Cho, Y., & Jiang, Y. (2025). AI applications for chronic condition self-management: Scoping review. *Journal of Medical Internet Research*, 27, e59632.
- [8] Du, Y. et al. (2025). Artificial intelligence in chronic disease self-management: Current applications and future directions. *Frontiers in Public Health*, 13, 1689911.
- [9] Reinhold, A. (2026). Memory as safety infrastructure: Evaluating provenance-aware persistent memory architectures for stateful LLM systems. Zenodo. DOI: 10.5281/zenodo.19244253.
- [10] Reinhold, A. (2026). Circular associative memory architecture: A framework for emotionally-keyed AI memory systems. Zenodo. DOI: 10.5281/zenodo.19051834.
- [11] Vayena, E. & Blasimme, A. (2018). Health research with big data: Time for systemic oversight. *The Journal of Law, Medicine & Ethics*, 46(1), 119–129.
- [12] Wagner, E.H. et al. (2001). Improving chronic illness care: Translating evidence into action. *Health Affairs*, 20(6), 64–78.
- [13] Vizient Inc. (2026). The chronic care reckoning: Redesigning access for an aging, complex population.
- [14] Packer, C. et al. (2023). MemGPT: Towards LLMs as operating systems. arXiv preprint arXiv:2310.08560.

Appendix A: Series Overview

This paper is part of the **Applied Persistent Memory Series**, published by Lorien’s Library LLC:

Paper 1: Persistent Memory as Mission-Critical Infrastructure for Long-Duration Spaceflight (DOI: 10.5281/zenodo.19257809)

Paper 2: Memory-Aware AI Systems for Permanent Lunar and Martian Habitation (DOI: 10.5281/zenodo.19260574)

Paper 3 (this paper): Provenance-Aware Memory Architecture for Chronic Healthcare Continuity

Paper 4: Haven: Music-Mediated Emotional Continuity for Veteran Reintegration

Open-source implementation: <https://github.com/LoriensLibrary/cama>

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